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AI for Bharat Hackathon

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Team Name : Tech Tapri

Team Leader Name : Aman

Problem Statement : **AI for Learning & Developer Productivity**

Brief about the Idea:

Career Coach – Autonomous Skill Gap Intelligence Engine

Career Coach is an agentic AI system that continuously analyzes a developer's real skill depth (via resume and GitHub), compares it against live job market demand, and autonomously re-prioritizes what the developer should learn next to maximize employability.

Unlike static learning platforms, Career Coach does not generate fixed roadmaps. It dynamically recalculates a Skill Impact Score using market demand, gap severity, career relevance, and user mastery level.

The system operates in a continuous AI loop:

Diagnose → Compare → Rank → Generate → Evaluate → Update → Re-rank

It pulls live job listings to understand real-time skill demand, evaluates actual project implementations from GitHub, updates mastery levels, and adjusts the learning roadmap automatically.

The result is a continuously evolving, market-aligned, performance-aware learning system that ensures developers learn the right skills at the right time.

Explain the following:

- **How different is it from existing ideas?**
- • Most learning platforms provide static courses and fixed roadmaps.
- • Career Coach continuously recalculates what to learn next based on live job market demand and actual GitHub performance.
- • It measures real implementation depth — not just course completion.
- • It autonomously reorders priorities without manual reset.
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- **How will it solve the problem?**
- • Extracts real skills from resume + GitHub.
- • Analyzes live job listings to detect market demand.
- • Computes a dynamic Skill Impact Score to rank gaps.
- • Generates personalized, duration-aware roadmaps.
- • Evaluates submitted projects and updates mastery levels.
- • Re-prioritizes learning based on performance feedback.
- Developers learn the right skills at the right time.
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- **USP of the Proposed Solution**
- • Autonomous skill gap intelligence engine
- • Live market-aligned roadmap generation
- • Real GitHub-based skill evaluation
- • Continuous AI-driven re-prioritization
- • Impact-driven learning, not template-driven learning

List of Features Offered by Career Coach:

1. AI Skill Extraction

- Resume parsing (PDF → structured skills)
- GitHub repository analysis
- Real implementation depth detection

2. Live Market Intelligence Layer

- Pulls live job listings
- Extracts skill demand using AI
- Updates market demand weights dynamically
- Weekly demand refresh

3. Skill Impact Scoring Engine (Core Innovation)

- Calculates dynamic Skill Impact Score
- Ranks skills by employability value
- Continuously re-prioritizes learning path

4. Personalized Adaptive Roadmap

- User selects target role
- User defines duration & weekly hours
- AI adjusts depth & complexity
- Roadmap auto-updates based on performance

5. Intelligent Project Evaluation

- GitHub project submission
- Code completeness analysis
- Best practice detection
- Concept mastery scoring
- Automatic mastery level update

6. Agentic Learning Loop

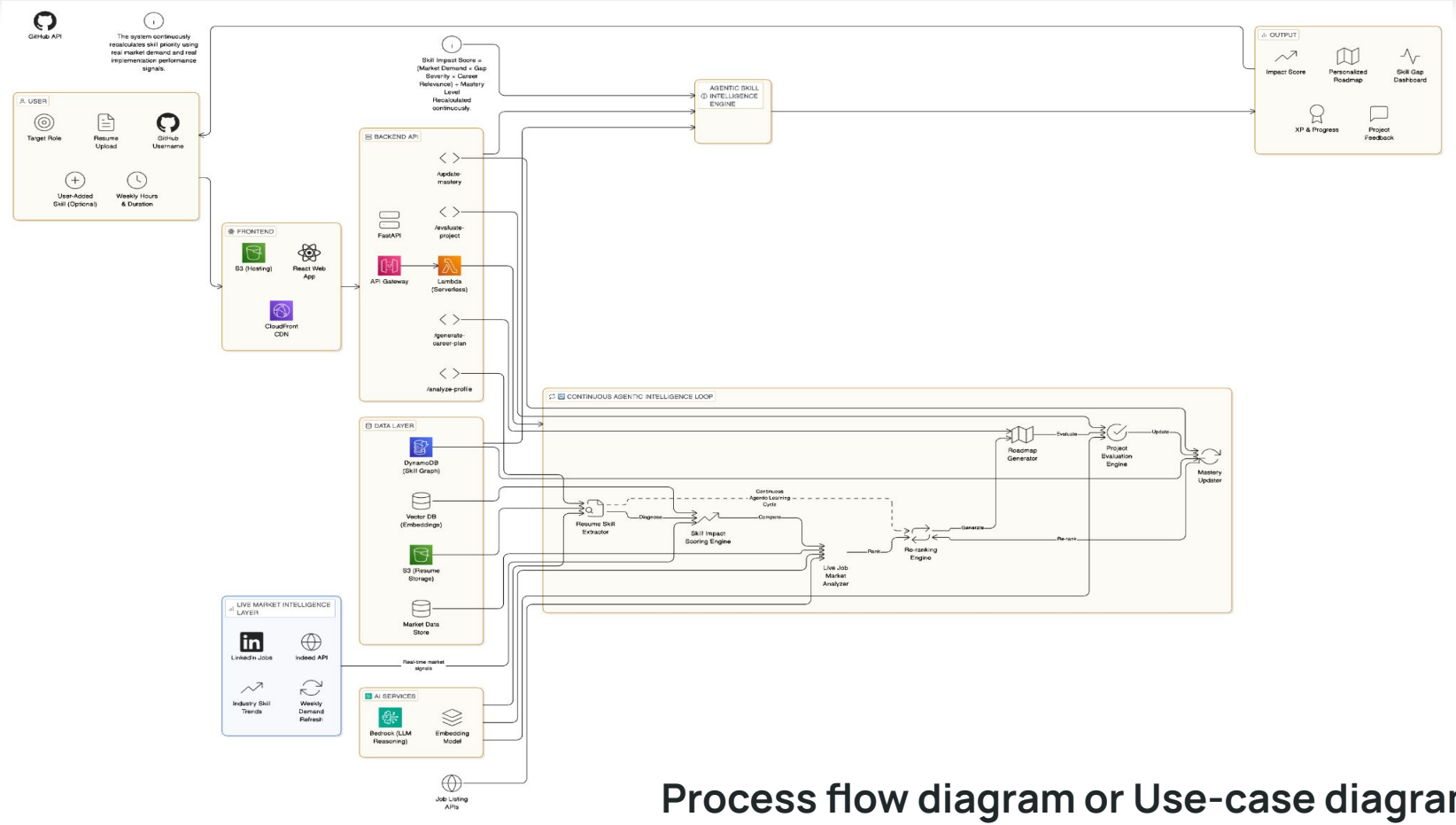
Diagnose → Compare → Rank → Generate → Evaluate → Update → Re-rank
Continuous autonomous adaptation.

7. Smart Gamification

- XP based on real mastery improvement
- Performance-based leveling
- Community access unlocks

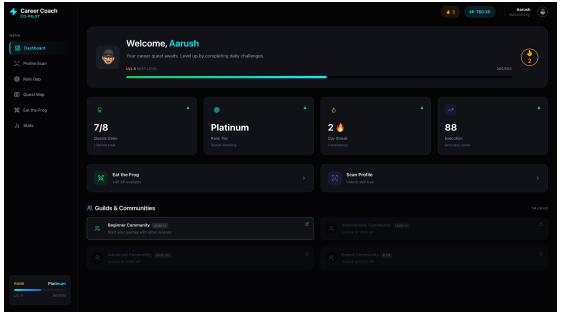
8. Multi-Segment Support

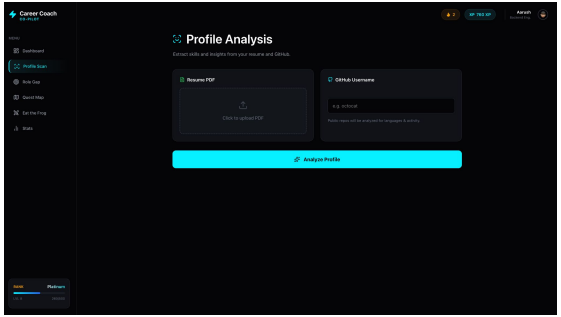
- University students
- Self-learners
- Working professionals
- Institutional licensing model

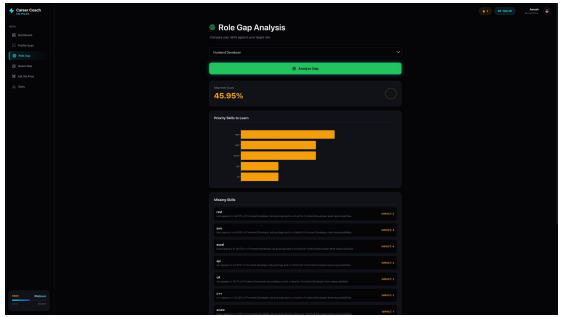


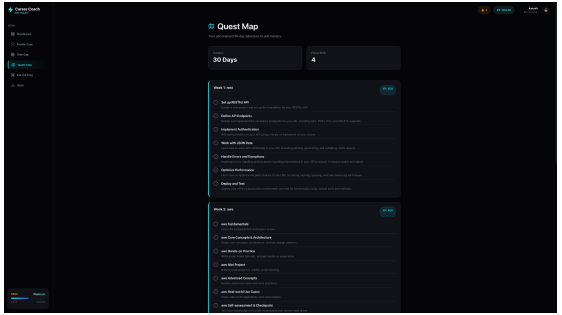
Process flow diagram or Use-case diagram

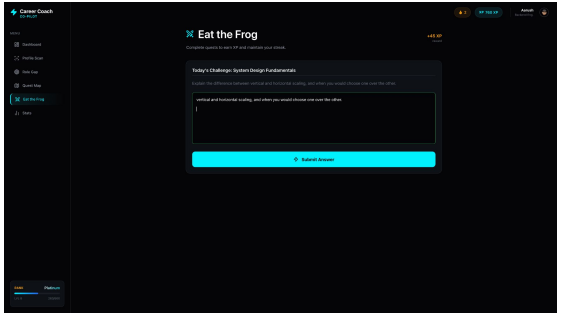
Wireframes/Mock diagrams of the proposed solution:



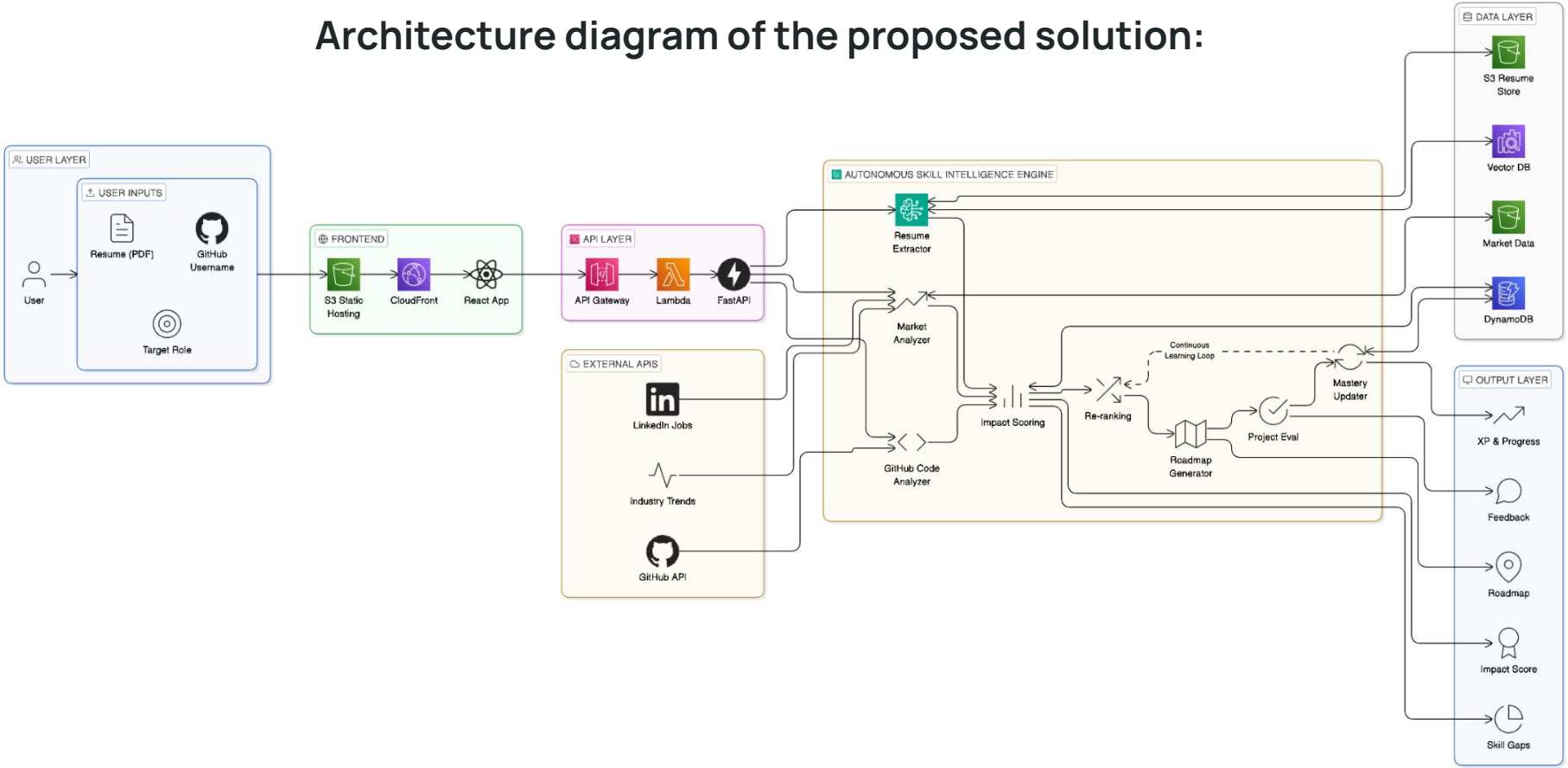








Architecture diagram of the proposed solution:



Technologies to be Used in the Solution:

Cloud Infrastructure (AWS)

- Amazon S3 – Static frontend hosting
- Amazon CloudFront – Global CDN
- Amazon API Gateway – Secure API routing
- AWS Lambda – Serverless backend execution
- Amazon DynamoDB – Skill graph & mastery storage
- Amazon S3 – Resume & document storage
- Amazon Bedrock – LLM-powered reasoning
- Amazon CloudWatch – Monitoring & logging

AI & Intelligence Layer

- Large Language Models (via Amazon Bedrock)
- Embedding Models for semantic skill matching
- Vector Database for skill similarity search
- Agentic Re-ranking Engine
- Dynamic Skill Impact Scoring Engine

Intelligence Capabilities

- Resume parsing (NLP-based skill extraction)
- GitHub repository analysis
- Live job listing analysis
- Semantic skill similarity computation
- Continuous mastery recalibration

External Integrations

- GitHub API
- Job Listing APIs (LinkedIn / Indeed)
- Market trend signals

Frontend & Backend

- React (Frontend UI)
- FastAPI (Backend framework)
- Python (AI logic & orchestration)

Security & Scalability

- Serverless architecture
- Role-based access control
- Encrypted storage
- Auto-scaling compute

Estimated Implementation Cost (MVP Stage):

Career Coach is designed using a serverless AWS architecture to keep costs low and scalable.

Estimated Monthly Cost (Hackathon / Early MVP Phase):

- AWS Lambda (compute): $20 – $40
- API Gateway: $10 – $20
- DynamoDB (Skill Graph & Mastery Data): $15 – $30
- Amazon S3 (resume storage + frontend hosting): $5 – $10
- CloudFront CDN: $5 – $15
- Amazon Bedrock (LLM inference): $50 – $150 (usage-based)
- Vector Database (embeddings storage): $20 – $40

Total Estimated Monthly Cost:
~ $120 – $300 per month

Scalability Model:

- Fully serverless → scales automatically with user growth
- Cost grows proportionally with usage
- Estimated cost per active user at scale: $0.50 – $2/month

Cost Optimization Strategy:

- Use embeddings to reduce repeated LLM calls
- Cache skill impact scores
- Batch-update job market data weekly
- Serverless auto-scaling to avoid idle infrastructure cost

Business Sustainability
Freemium Model:

- Free basic skill analysis
- Pro plan ($10–20/month) for live market intelligence & advanced evaluation
- University & corporate licensing

Infrastructure cost is sustainable and profitable at scale.

Unit Economics Example:

Example at scale (1,000 Pro users):

- Revenue: 1,000 × $15 = $15,000/month
- Estimated Infra Cost: ~$2,000/month
- Gross Margin: ~80%+

Security & Responsible AI:

- Resume data encrypted at rest (S3 + IAM policies)
- Secure API access (JWT/Auth layer)
- GitHub OAuth (no password storage)
- User data not sold or shared

Alignment with Hackathon Requirements:

Meaningful Use of AI (Not Rule-Based)

- Uses LLMs to extract skills from unstructured resumes & GitHub repositories
- Uses embeddings to compute semantic skill similarity
- AI-driven Skill Impact Scoring Engine dynamically ranks learning priorities
- Intelligent project evaluation using code understanding models
- Autonomous roadmap re-ranking based on market signals & performance

AI is required for reasoning, interpretation, and adaptive decision-making.

Improves Learning & Developer Productivity

- Prevents learners from studying outdated or low-impact skills
- Aligns learning with real-time job market demand
- Converts skill gaps into structured, time-bound learning paths
- Automatically evaluates projects and provides improvement feedback
- Reduces guesswork in career growth

Learning becomes measurable, adaptive, and outcome-driven.

Designed for Bharat

- Helps students from Tier 2 & Tier 3 colleges align with industry demand
- Reduces dependency on expensive career coaching
- Enables industry-aligned skill development at scale
- Can be deployed across universities for standardized upskilling

Built on Scalable AWS Infrastructure

- Serverless architecture for low cost & high scalability
- Amazon Bedrock for AI reasoning
- Secure storage & privacy-first design
- Auto-scaling infrastructure ready for nationwide deployment

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Thank You